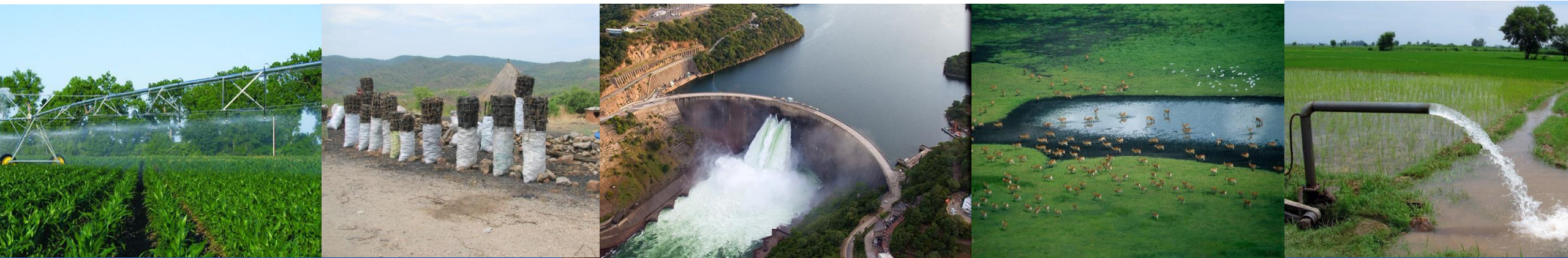


Exercise 11: Structure of CWatM

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Structure of CWatM – Exercise 11

Change the output of CWatM

1. Model structure
2. Model principles
3. Data management
4. Hydrological procedures
5. GitHub



Structure of CWatM – Exercise 11

Model structure of CWatM

- not only for programmers
- but also to understand how CWatM is working



Please also have a look at:

<https://github.com/iiasa/CWatM>

<https://cwatm.iiasa.ac.at/sourcecode.html>

Model principles 1:

- splitting between management tools

(reading, writing files , error messages, managing time, reading the settings)

➡ in `cwatm\management_modules`

- and hydrological modules

(calculation of evapotranspiration, soil, routing, ...)

➡ in `cwatm/hydrological_modules`

Model principles 2:

Hydrological modules

- Splitting between initial phase
 - everything which is needed for initializing a process
 - e.g. loading maps which are not changing during run
- and dynamic phase
 - everything which runs in time steps

Model principles 3:

Parameter and option are separated from the source code by a Settings file:

- Option can include different methods (on/off)
 - evaporation(Penman Monteith vs. Priestley-Taylor)
 - snow (day-degree vs. day-degree and radiation)
- Section order can be changed!

Model principles 4:

All input maps can be of different format:

- spatial maps can be netCDF, geotif, or PCRaster .map
- some input can be a number or a map
- exception: spatial timeseries can only be netCDF

Model principles 5:

All global variables can be output in different format

- as timeseries for stations
- as netCDF spatial, spatial timeseries

Output variables:

<https://cwatm.iiasa.ac.at/listVariables.html#rst-variables>

Data management

- Data management
 - data_handling module
 - timestep module
 - configuration module
 - management_modules.messages module
 - management_modules.output module
 - management_modules.checks module
- Program management
 - globals module
 - dynamicModel module
 - replace_pcr module

Hydrological modeling

◦ Initialize

- miscInitial module
- initcondition module
- landcoverType module

◦ Hydrology I - from rain to soil

- readmeteo module
- inflow module
- snow_frost module
- evaporationPot module
- evaporation module
- interception module
- sealed_water module

◦ Hydrology II - from soil to river

- soil module
- capillarRise module
- groundwater module
- runoff_concentration module

◦ Hydrology III - Socio-economic - Water demand

- water_demand.water_demand
- water_demand.domestic
- water_demand.industry
- water_demand.livestock
- water_demand.irrigation
- water_demand.environmental_need

◦ Hydrology IV - Lakes, reservoirs and river

- lakes_reservoirs module
- lakes_res_small module
- routing_reservoirs.routing_kinematic module
- routing_reservoirs.routing_sub module

Structure of CWatM – Exercise 11

GitHub

Our software is open source.

We use a public repository to store our software – Github

<https://github.com/iiasa/CWatM>

As CWatM is a living software, we add things, change things, remove bugs
And we put our changes alive on GitHub



<https://github.com/>

<https://docs.github.com/en/free-pro-team@latest/github/getting-started-with-c>

<https://www.wikihow.com/Create-an-Account-on-GitHub>



We have a video explaining Github on Sharepoint:

[SO2/workshop_sofia_12_2024/github](#)

or [tutorial/exercise11/de_Bruijn_explaining_Github.mp4](#)

Structure of CWatM – Exerc

CWatM - GitHub

The screenshot shows the GitHub repository for CWatM. At the top, it indicates the current branch is 'develop', which is 4 branches and 9 tags away from 'main'. The repository is owned by 'mikhailsmilovic' and has 658 commits. The file list includes folders like 'Toolkit', 'Tutorials', 'cwatm', and 'mf6', and files like '.gitignore', 'LICENSE', 'README.md', 'report_pytest_111021.html', 'requirements.txt', 'run_cwatm.py', 'setup.py', and 'test_script.py'. Below the file list, there are tabs for 'README' and 'License'. The 'README' tab is active, showing the title 'Community Water Model (CWatM)'. Below the title, there are badges for 'last commit' (july), 'license' (GPL-3.0), 'python' (3.7 | 3.8 | 3.9 | 3.10 | 3.11), 'pytest' (passing), 'codecov' (88%), 'repo size' (241.2 MiB), 'docs' (passing), and 'DOI' (10.5281/zenodo.3528097). The README text includes links to the user manual and model documentation at <https://cwatm.iiasa.ac.at>, a link to the GitHub forum for questions, a link to CWatM tutorials on YouTube, and a link to the CWatM-Earth-30min repository for input data.

develop 4 Branches 9 Tags Go to file Code

This branch is 11 commits ahead of, 4 commits behind main

mikhailsmilovic crop-spec. trans. and outputs cbf5a9d · last week 658 Commits

Toolkit	Updated .rst pages publication.rst and setup.rst with changed d...	2 months ago
Tutorials	Calibration fix	last year
cwatm	crop-spec. trans. and outputs	last week
mf6	Including modflow	last year
.gitignore	Open water evap updated	last year
LICENSE	Updating cwatm to work from pip install and manual install	4 years ago
README.md	Update README.md	last year
report_pytest_111021.html	Change: Merge branch Luca-CWatM_ModFlow6-varsteps	3 years ago
requirements.txt	Flopy version in requirements and Burgenland option in landc...	3 years ago
run_cwatm.py	Bugfixes	last year
setup.py	chg: turial 7, travis	4 years ago
test_script.py	Removing load from settings not used	last year

README License

Community Water Model (CWatM)

last commit july license GPL-3.0 python 3.7 | 3.8 | 3.9 | 3.10 | 3.11 pytest passing codecov 88% repo size 241.2 MiB docs passing DOI 10.5281/zenodo.3528097

User manual and model documentation at <https://cwatm.iiasa.ac.at>.

Questions? Start a discussion on our [GitHub forum](#) and check out our [CWatM tutorials on YouTube](#).

Our repository [CWatM-Earth-30min](#) contains input data for CWatM at 30 arcminutes and further links to climate and higher resolution input data.